



Energy Division

RPIT

Raychem plug in termination system
up to 42 kV for gas insulated switchgear

RPIT

Features

- Separable inline connection for high current, gas insulated switchgears up to 42 kV
- Termination mates interfaces in accordance to EN50180 and EN50181 for inner cone connections
- The contact parts are designed for stranded circular aluminium or copper conductors in accordance to IEC-60228

Technical Information

The increasing popularity of gas insulated switchgear called for the development of appropriate connection systems to standardised bushings in accordance to EN50181.

Tyco Electronics Energy Division has developed two systems, one is related to the outer cone and the other one to the inner cone system. Later is used in power switchgear and power transformers.

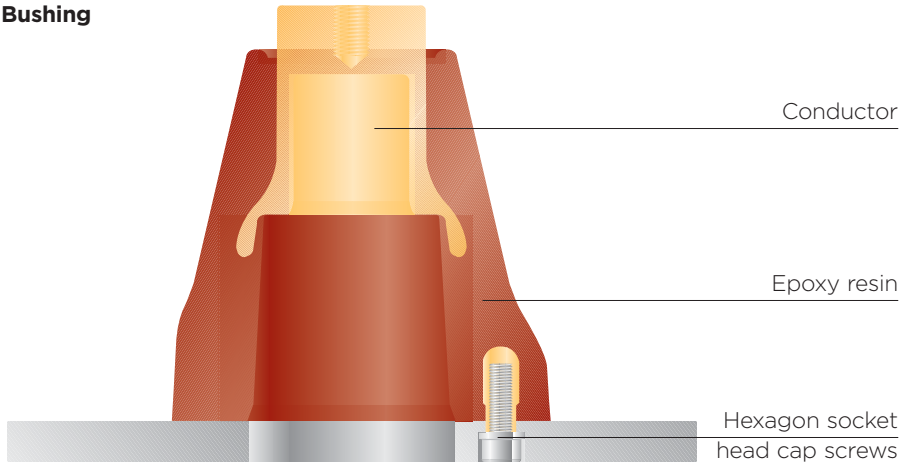
Tyco Electronics Energy Division has several decades of experience in the field of hermetically insulated termination systems for medium voltage applications. Thus the bushings designed for gas insulated switchgear meet the standard EN50181 for connection type size 2 (800 A) and size 3 (1250 A) at operating voltage 12 kV up to 36 (42) kV.

The basic construction of the mating termination follows well known design principles and incorporates a lamellate contact system which ensures reliable current transmission via a cone clamp to the conductor. The interface between silicone body of the termination and the bushing is kept permanently sealed via the pressure of a spring between the insulator and the flange of the termination cover.

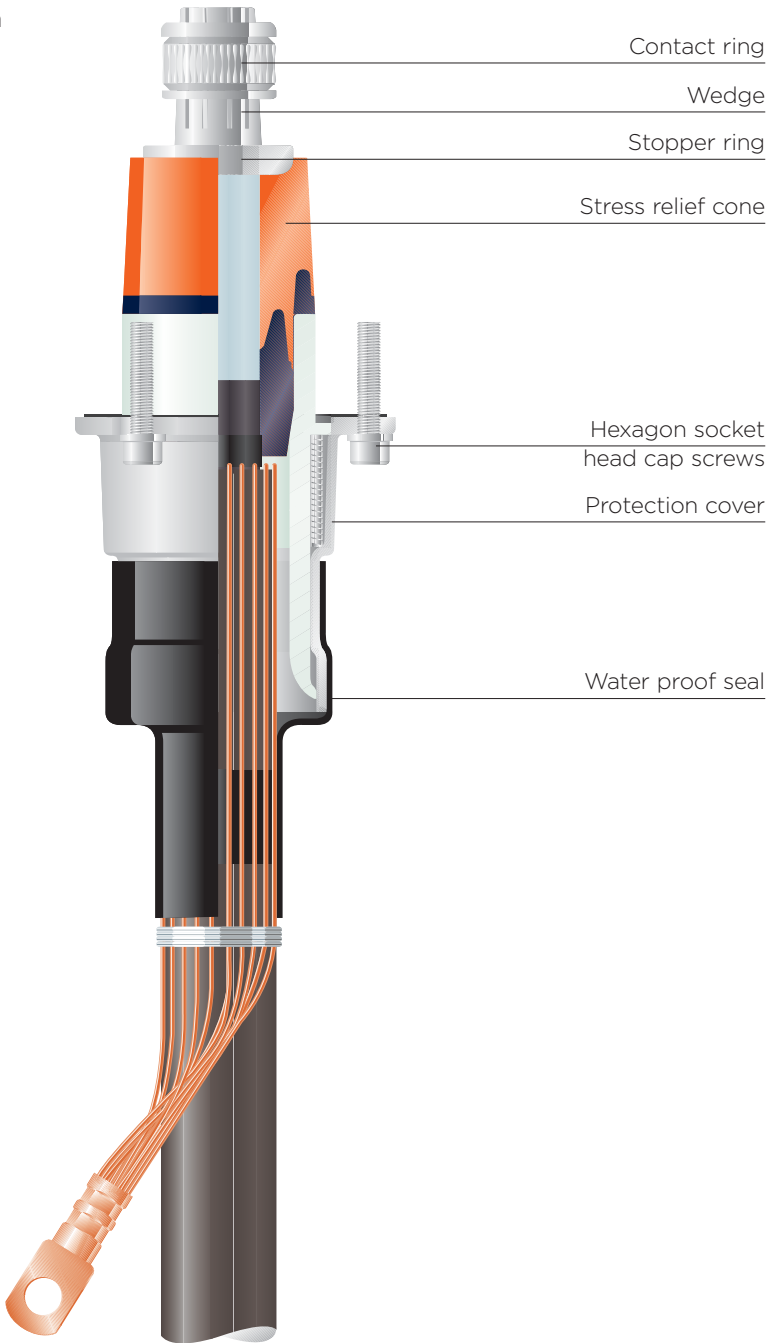
The termination cover is sealed at the flange area with well known sealing methods and at the cable side with heat-shrinkable tubing against ingress of moisture.

Raychem plug in termination system up to 42 kV
for gas insulated switchgear
Nominal current up to 1250 A

Bushing



Termination



Testing methodes according to CENELEC HD629.1 S2

Product size	2	2	3	
Cross section (mm²)	300	185	630	
Diameter over insulation (mm)	max. 36.6	max. 36.6	max. 50.0	
	Testing values for rated voltage			Results
U ₀ / U	12.7 / 22	19 / 33	20.8 / 36	
U _m	24	36	42	
DC withstand voltage 15 min (kV)	76	114	125	no breakdown or flashover
AC withstand voltage 5 min (kV)	57	85.5	93.5	no breakdown or flashover
PD at ambient temperature (kV)	< 2 pC	< 2 pC	< 2 pC	max. 10 pC at 2 U ₀
Impulse voltage at elevated temperature (kV)	150	200	200	10 impulses of each polarity; No breakdown
Heating cycle voltage in air (kV)	32	48	52	63 cycles at 2.5 U ₀ ; no breakdown
Heating cycle voltage in water (kV)	32	48	52	63 cycles at 2.5 U ₀ ; no breakdown
Thermal short circuit (conductor) (kA)	31.5 / 3 sec	31.5 / 1 sec	50 / 5 sec	2 short circuits to raise conductor to 250°C; no breakdown
Dynamic short circuit (kA)	125	125	125	1 short circuit at I _d ; no breakdown
Disconnection / Connection	5 x pass	5 x pass	5 x pass	5 complete operations; no visible damage to contact
Partial discharge at elevated and ambient temperature (kV)	< 2 pC	< 2 pC	< 2 pC	max. 10 pC at 2 U ₀
Impulse voltage at ambient temperature (kV)	150	200	200	10 impulses of each polarity; no breakdown
AC withstand voltage 15 min (kV)	32	48	52	no breakdown or flashover

Cable terminations

Type	Size	Nominal current (A)	System voltage (kV)	Cross section (mm²)	Dia over conductor (mm)	Dia over insulation (mm)
RPIT-321x	2	800	12	95 - 300	11.0 - 21.6	19.3 - 30.4
RPIT-521x	2	800	24	50 - 300	7.7 - 21.6	20.2 - 34.6
RPIT-621x	2	630	36	50 - 185	7.7 - 16.8	25.2 - 35.1
RPIT-331x	3	1250	12	240 - 630	17.8 - 32.5	26.4 - 29.6
RPIT-531x	3	1250	24	150 - 630	13.9 - 32.5	26.5 - 45.6
RPIT-631x	3	1250	36 / 42	95 - 630	11.0 - 32.5	28.5 - 49.2

Note: Product is available with and without test point



Bushing Parts

Size	Modification	Reference number
Size 2 (800A)	without test point	RPIS-2-IS
Size 2 (800A)	with test point	RPIS-2-IS-VD
Size 3 (1250A)	without test point	RPIS-3-IS
Size 3 (1250A)	with test point	RPIS-3-IS-VD



Insulating cap

Size	Reference number
Size 2	RPIC-2
Size 3	RPIC-3



Compression tool

Metal tool case including an easy to handle compression tool for wedge and contact ring and all necessary adaptation rings.

Reference number: IT-1000-025



Parts to be used for compression of contact elements



Parts to be used for removal of the contact elements



Insert lever and support chain

Description	Reference number
Insert lever	IT-1000-026
Support chain	IT-1000-027



Impact device

Description	Reference number
Device for size 2	IT-1000-028
Device for size 3	IT-1000-029

All of the above information, including drawings, illustrations and graphic designs, reflects our present understanding and is to the best of our knowledge and belief correct and reliable. Users, however, should independently evaluate the suitability of each product for the desired application. Under no circumstances does this constitute an assurance of any particular quality or performance. Such an assurance is only provided in the context of our product specifications or explicit contractual arrangements. Our liability for these products is set forth in our standard terms and conditions of sale. Raychem, TE Logo and Tyco Electronics are trademarks.

Energy Division – economical solutions for the electrical power industry: cable accessories, connectors & fittings, electrical equipment, instruments, lighting controls, insulators & insulation enhancement and surge arresters.

Tyco Electronics Raychem GmbH
Energy Division
Finsinger Feld 1,
85521 Ottobrunn/Munich, Germany

Phone: +49-89-6089-0
Fax: +49-89-6096345

<http://energy.tycoelectronics.com>